William Marella

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Senior data scientist, ML engineer, and serial builder with expertise in causal inference, experimentation, and applied ML. Background spans end-to-end ML systems, statistical modeling at scale, and full-stack engineering. Seeking ML/Applied Science or senior data science roles focused on experimentation, ML, or causal reasoning.

EDUCATION

University of Cambridge November 2024
M. Phil. Population Health Sciences, Specialization in Health Data Science

Brown University May 2023
B.A. Biology, GPA: 3.96

WORK EXPERIENCE

Senior Data Scientist New York, New York
Columbia University, Belsky Lab *September 2024 - Present*

- Analytical and engineering lead for Columbia's GeroScience Computational Core. Partnered with faculty across Columbia to design and execute causal inference, ML, and predictive modeling projects in biological aging and longevity science
- Mentored master's students in biostatistics and data science

ARPA-H FAST Grant | *R, Python, Causal Inference, ML*

- Led design and execution of causal inference and ML strategy for an awarded **$10M ARPA-H grant**. Architect and lead developer for statistical R packages used internally and externally by collaborators including **Novo Nordisk and Lilly**
- Developed and implemented a **full causal mediation analysis** to decompose total intervention effects into direct and indirect pathways through multi-omic intermediates
- Built and maintain ML pipelines predicting clinical endpoints from high-dimensional omics data; designed to handle repeated-measures structure and collaborator-facing reproducibility requirements

FraminghamPACE | *R, Hierarchical Modeling, ML*

- Developed a hierarchical model of the pace of aging and machine learning training methods to create a **state-of-the-art epigenetic aging clock** with improved predictive performance, technical reliability, and responsiveness to intervention
- **Inventor, commercially licensed algorithm to TruDiagnostic; patent pending, manuscript in preparation**

UK Biobank BioScore | *R*

- Collaborated with the founder of the HOOKE Longevity Clinic to construct a **novel composite aging score in the UK Biobank (500,000+ participants),** independently designing and implementing the measure
- Validated BioScore across retrospective, cross-sectional, and prospective outcomes including time-to-event analyses. **Outperformed PhenoAge, the current field standard. Manuscript in preparation**

Visiting Researcher Cambridge, UK
University of Cambridge, Inouye Lab *June 2024 - Present*

- Developed scalable Bayesian modeling of disease comorbidity structure in UK Biobank data
- Wrote Python libraries for reliable statistical inference, including convergence diagnostics and multi-threaded MCMC coordination

Latent Factor Allocation (LFA) | *Python, Probability* **|** *GitHub*

- Applied unsupervised Bayesian modeling and scalable inference to identify latent disease groupings from comorbidity data
- **Derived mean-field variational inference and partially collapsed Gibbs sampling algorithms**. Benchmarked accuracy and time complexity on **simulated data** designed to mirror real UK Biobank data
- Implemented both methods from scratch in Python (NumPy/SciPy) with convergence diagnostics and multi-threaded MCMC chain coordination. **Packaged as a public Python library**

PUBLICATIONS AND PATENTS

"An Epigenetic Speedometer to Measure Pace of Aging: FraminghamPACE" | **First author**. Manuscript in preparation.
"*UKB BioScore: A Data-Driven Clinical Measure of Biological Aging in the UK Biobank*" | **First author**. Manuscript in preparation.
"FraminghamPACE" | **Inventor**. Patent application in preparation. Columbia University Technology Ventures. **Commercially licensed to TruDiagnostic**.

PRESENTATIONS

NIA Biomarker Meeting 2026 | St. Louis, MO **|** Invited Oral Presentation **|** *FraminghamPACE: A New DNA Methylation Biomarker of the Pace of Aging*

Biomarkers of Aging Consortium 2025 | Boston, MA | Poster Presentation | *FraminghamPACE: An Epigenetic Speedometer of the Pace of Aging*

SELECTED PROJECTS

QueryGaP | *Python, PostgreSQL, JavaScript* | *Website*

- **Scraped and indexed all dbGaP and UK Biobank variable documentation** into PostgreSQL with hybrid semantic and keyword search (pgvector, tsvector) to drastically **speed up navigation of thousands of phenotype variables**
- Built an AI agent to query the database and return natural language answers about study documentation, **cutting search time by over 95%**
- **Used daily at Columbia's Belsky Lab.** Growing user base across Columbia and external research institutions

ASCII Doodle | *Python, PyTorch, Transformers* **|** *GitHub*, *HuggingFace*

- Text-to-ASCII art generator powered by a **300M parameter Diffusion Transformer (DiT)** trained with flow matching over a **VAE-compressed latent space** of 32x16 ASCII token grids
- Trained on Quick, Draw! using **Modal A100s**, weights on HuggingFace
- Conditioned on **CLIP text embeddings** with classifier-free guidance for controllable prompt fidelity. Generalizes to synonyms and near-out-of-distribution prompts unseen during training

OpenCode Protect | *TypeScript, Node.js, Bun* | *GitHub*

- Forked OpenCode to implement **kernel-level isolation of AI agent commands,** running them as a restricted Unix user to protect sensitive credentials and files
- Built to address **prompt injection vulnerabilities** that trivially bypass prompt-level protections in existing coding agents

CostcoGuessr | *TypeScript, Next.js, Bun, PostgreSQL* | *Website*

- Scraped **10,000+ Costco products across 11 countries**, stored in PostgreSQL with images on Vercel Blob
- Built a full-stack daily price-guessing game in TypeScript/Bun, deployed via Railway and Vercel, reaching **200 daily users**

Open-Source Contributions: *TypeScript, React, Bun, Node* | *PR#1*, *PR#2*

- Merged navigation and UI features and fixes into **OpenCode** and **PostHog** codebases

SKILLS

Languages: Python, R, SQL, TypeScript, JavaScript, Rust, Bash, Unix/Linux
Frameworks & Libraries: PyTorch, scikit-learn, HuggingFace, LangChain/LangGraph, NumPy, SciPy, pandas, Optuna, React, tidyverse, lme4, emmeans, survival, limma, requests, BeautifulSoup, matplotlib, seaborn, ggplot2
Infrastructure & Tools: Git, GitHub, PostgreSQL, pgvector, Docker, Modal, Vercel, Railway, PostHog
AI & LLM Ecosystem: OpenAI, Anthropic, Google Gemini, Mistral, xAI; Claude Code, Codex, Cursor, OpenCode, AntiGravity, Google AI Studio; prompt engineering, context engineering, multi-agent orchestration, agentic AI, RAG pipelines, vector search, LLM evaluation & benchmarking, LLM-as-a-judge, simulation and synthetic data generation
Experimentation & Causal Inference: A/B testing, difference-in-differences, causal mediation analysis, treatment effect estimation, quasi-experimental design, propensity score weighting, sensitivity analysis, hypothesis testing
Statistical Methods: Mixed-effects models, survival analysis, longitudinal data analysis, hierarchical models, Bayesian inference (MCMC, variational inference), multiple testing & FDR correction, power analysis, parametric & non-parametric significance testing
Machine Learning: Supervised & unsupervised learning, regularized regression, gradient boosting (XGBoost, LightGBM, CatBoost), ensemble methods, dimensionality reduction, clustering, NLP & text mining, feature selection, hyperparameter optimization, model interpretability (SHAP), double/debiased ML, neural networks (MLP, ViT, DiT), diffusion models, flow matching, CLIP